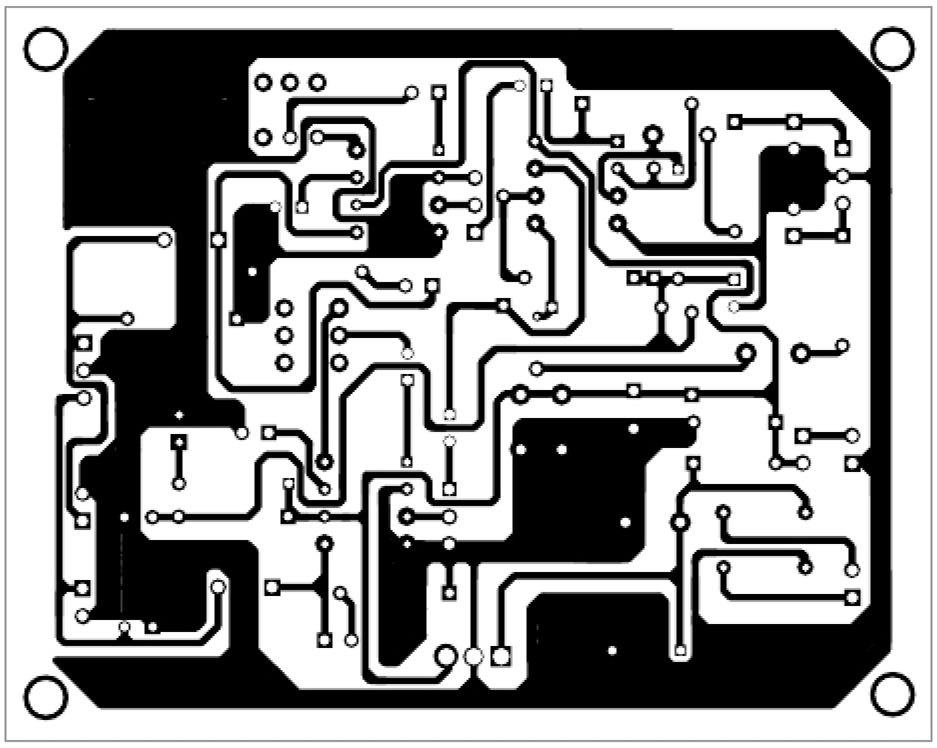


Fig. 2: Actual-size PCB layout of universal audio signal tracer and tester

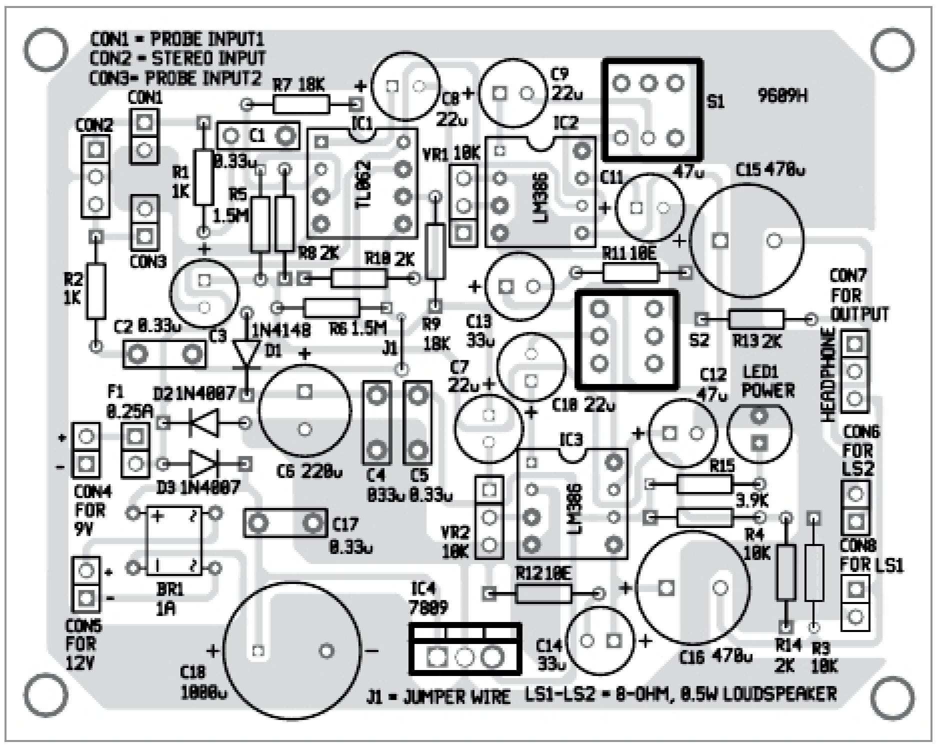


Fig. 3: Components layout for the PCB

attention to the maximum output power so as to not damage the headphones and your ears. Volume of both the channels is adjusted independently with potentiometers VR1 and VR2, respectively.

Power supply. A 9V DC power supply at CON4 is used to power the circuit. A 12V to 20V AC or DC power supply can also be used at CON5. Bridge rectifier BR1 and voltage regulator 7809 are used to convert AC input to 9V DC output. If power supply is available at CON4, do not apply any voltage at CON5.

The circuit will also work with a 12V dry or rechargeable battery connected to CON5. The power supply range decides the output power of LM386.

Construction and testing

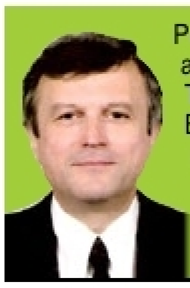
An actual-size PCB layout of the universal audio signal tracer is shown in Fig. 2 and its components layout in Fig. 3. Assemble the components as per the circuit diagram on the PCB. Connect 9V DC to the circuit. The glowing of LED1 indicates the presence of a power supply in the circuit.

Connect any of the inputs (CON1-CON3) to an audio source. Adjust VR1 and VR2 until you get comfortable sound levels at outputs (loudspeaker or headphones).

To use as simple audio signal tracer, connect any of the loudspeakers at CON6 or CON8, or headphones at CON7. Then, connect an audio probe to either CON1 or CON3. You can also use a stereo audio jack at CON2 and headphones at CON7 to perform various tests and troubleshooting in audio equipment. EFY

PARTS LIST

Semiconductors:	
IC1	- TL062 operational amplifier
IC2-IC3	- LM386 audio amplifier
IC4	- 7809, 9V voltage regulator
BR1	- 1A bridge rectifier
D1	- 1N4148 signal diode
D2-D3	- 1N4007 rectifier diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1-R2	- 1-kilo-ohm
R3-R4	- 10-kilo-ohm
R5-R6	- 1.5-mega-ohm
R7, R9	- 18-kilo-ohm
R8, R10, R13-R14	- 2-kilo-ohm
R11-R12	- 10-ohm
R15	- 3.9-kilo-ohm
VR1-VR2	- 10-kilo-ohm potmeter
Capacitors:	
C1-C2, C4-C5, C17	- 0.33µF ceramic disk
C3	- 100µF, 25V electrolytic
C6	- 220µF, 25V electrolytic
C7-C10	- 22µF, 25V electrolytic
C11-C12	- 47µF, 25V electrolytic
C13-C14	- 33µF, 25V electrolytic
C15-C16	- 470µF, 25V electrolytic
C18	- 1000µF, 35V electrolytic
Miscellaneous:	
S1-S2	- On/off switch
F1	- 0.25A fuse
CON1, CON3, CON6, CON8	- 2-pin connector
CON2, CON7	- 3-pin connector for stereo input probe
CON4-CON5	- 2-pin connector
LS1, LS2	- 8-ohm, 0.5W loudspeaker
	- Audio input probe
	- Stereo headphones
	- 9V DC or 12V AC/DC power supply



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